

JAIA Caribvoices Competition Documentation

1. Overview & Objectives

This solution develops an Automatic Speech Recognition (ASR) model for Caribbean dialects using fine-tuning of a large-scale pretrained model. The primary objective was to reduce Word Error Rate (WER) when transcribing audio containing Caribbean linguistic patterns. The final system fine-tunes OpenAI's Whisper-Large-V3 model using a custom HuggingFace dataset constructed from the provided training audio and transcripts, the datasets and model are publicly available on huggingface.

2. Architecture Overview

The system is composed of three main components:

1. Data Processing

- Convert provided audio/transcription files into a unified HuggingFace dataset
- Perform train/validation split
- Push dataset to HuggingFace Hub

2. Model Fine-Tuning

- Load Whisper-Large-V3
- Train using a custom evaluation loop
- Evaluate using WER during validation

3. Inference / Prediction

- Load fine-tuned model

- Run inference on test audio
- Produce JSON/CSV prediction output for submission

Execution environment: Google Colab Pro with A100 GPU (80GB).

3. ETL (Data Processing)

Extract

- Provided inputs:
 - `train.csv` (audio filename + transcription)
 - `test.csv` (audio filename only)
 - Raw audio files
- Test set lacked transcription; a temporary empty column was added to create a consistent schema

Transform

- Mapped audio → transcript using the shared filename identifier
- Converted CSVs into a HuggingFace `Dataset`
- Cast the audio column using HF Audio type
- Performed programmatic train/validation split

Load

- Final dataset uploaded to: `neddamj/carib-voices_data` on HuggingFace hub
- Dataset used directly by the training notebook via HF streaming APIs

4. Data Modeling

Model

- Base model: **Whisper-Large-V3**
- Framework: PyTorch + HuggingFace Transformers

Hyperparameters

- Learning Rate = 1e-5
- Evaluation Metric = WER

Training

- Fine-tuned on train subset
 - Validation applied during training
 - Custom evaluation loop implemented using `evaluate.load("wer")` due to an issue in the built-in HF trainer evaluation, which stalled on the final step of evaluation
 - Uploaded fine-tuned model to [neddamj/whisper-large-carib](#) on huggingface
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5. Inference

Inference is executed via a standalone notebook. Steps:

1. Load fine-tuned model from HF checkpoint
2. Load test dataset
3. Generate transcriptions using `Whisper generate()`
4. Save results to submission file

Hardware: Google Colab A100 (80GB)
Time: slightly over 1 hour for full test set inference

6. Runtime

Stage	Hardware	Duration
Data preparation	CPU	~15-30 min
Fine-tuning (2,500 steps)	A100	~3.5 hours
Inference (full test)	A100	~1 hour+

7. Performance Metrics

- **Public leaderboard: 0.031381541**
 - **Private leaderboard: 0.030936289**
 - No additional internal metrics tracked
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8. Error Handling & Logging

Not explicitly implemented beyond standard exceptions and console output. Training and inference were monitored manually using Colab logs and printed evaluation metrics.

9. Maintenance & Monitoring

This prototype solution was built for a hackathon submission and does not implement long-term monitoring, data drift detection, or retraining pipelines. Future work could incorporate periodic retraining when new Caribbean dialect data becomes available.